STUDENT MARKS ANALYZER (IN-MEMORY)

C Programming Mini Project Report

Submitted By

Bhavani S – 25BCS211
Deva G – 25BCS212
Dhanya Sree V S – 25BCS213
Elangamanimukilan K – 25BCS214
Gowtham S – 25BCS215

Course: B.Sc Computer Science
Subject: C Programming Lab
Team: Team – 3

1. Introduction

In educational institutions, analyzing student marks is an important task to evaluate academic performance. Traditionally, teachers calculate total marks, averages, and grades manually, which can take a lot of time and effort.

The project titled 'Student Marks Analyzer (In-Memory)' is developed using the C programming language to automate the process of analyzing student marks. The program stores student data in memory using structures and arrays. It calculates the total marks, average marks, and grade for each student.

The system also identifies the top performer among students based on the highest total marks. Since the program stores data in memory during execution, it does not require external files or databases. This project demonstrates the use of fundamental programming concepts in C.

2. Objectives of the Project

The main objectives of this project include:

• To store student information using structures in C programming.

• To calculate total and average marks for each student.

• To assign grades based on average marks.

• To display results in a structured format.

• To identify the top-performing student.

• To practice programming concepts such as arrays, loops, conditions, and structures.

3. Problem Statement

In many educational institutions, calculating student results manually can be time-consuming and prone to errors. Teachers often need to calculate totals, averages, and grades for several students, which increases workload.

Existing digital systems may require complex databases or file management systems. For learning purposes and small-scale applications, a simple solution is required.

This project provides a simple console-based C program that accepts student marks, calculates total and average scores, assigns grades, and identifies the top performer.

4. System Requirements

Hardware Requirements

- Basic computer system
- Minimum 2 GB RAM
- Keyboard and monitor

Software Requirements

- Operating System: Windows / Linux
- Compiler: Turbo C / GCC / Dev C++
- Programming Language: C

5. System Design

The system uses structures and arrays to store student information and marks. A structure named 'Student' is defined to hold student details such as ID, name, marks, total, average, and grade.

The program reads input values, performs calculations, and displays results. The design is simple and suitable for learning fundamental programming techniques.

6. Module Description

Input Module

Collects student ID, name, and subject marks from the user.

Processing Module

Calculates total marks and average marks for each student.

Output Module

Displays results in a clear tabular format.

Top Performer Module

Identifies the student with the highest total marks.

7. Algorithm

Step 1: Start the program.

Step 2: Enter number of students.

Step 3: Enter number of subjects.

Step 4: Input student ID, name, and marks.

Step 5: Calculate total marks.

Step 6: Calculate average marks.

Step 7: Assign grades.

Step 8: Display results.

Step 9: Identify top performer.

Step 10: End program.

8. Implementation

The program uses C programming concepts such as structures, arrays, loops, and conditional statements. Structures help organize student data, arrays store marks for multiple subjects, loops process multiple students, and conditions assign grades.

9. Advantages

- Reduces manual calculation.
- Easy to understand and use.
- Demonstrates core programming concepts.
- Useful for academic practice.

Limitations

- Data is stored only temporarily in memory.
- No graphical interface.
- Not suitable for large databases.

10. Conclusion

The Student Marks Analyzer project demonstrates how C programming can automate the process of calculating and analyzing student marks. The program calculates totals, averages, assigns grades, and identifies the top performer. This project provides practical knowledge of programming concepts and can be expanded further with features like file storage and GUI.